

SARDAR PATEL UNIVERSITY
M. Sc. IT (Integrated) (Regular/NC) Examination – 9th Semester
PS09CHIT03: Compiler Design
Friday, 26th October, 2018

Time: 10:00 am to 1:00 pm

Max. Marks – 70

Q-1 MCQ

[08]

- 1) The analysis part is often called the _____ of the compiler.
A) Front end
B) Middle end
C) Back end
D) None of these
- 2) What is the output of lexical analyzer?
A) A parse tree
B) Intermediate code
C) A list of tokens
D) Machine code
- 3) Regular expression $(x/y)(x/y)$ denotes the set _____.
A) $\{xy, xy\}$
B) $\{x, y\}$
C) $\{xx, xy, yx, yy\}$
D) $\{x, y, xy\}$
- 4) In regular expressions, the operator '*' stands for _____.
A) Concatenation
B) Iteration
C) Selection
D) Addition
- 5) Grammar of the programming is checked at _____ phase of compiler.
A) Semantic analysis
B) Code optimization
C) Syntax analysis
D) Code generation
- 6) A grammar that produces more than one parse tree for same sentence is called as _____.
A) Ambiguous
B) Regular
C) Unambiguous
D) All of these
- 7) Which of the following field is not in triple?
A) Op
B) arg2
C) arg1
D) result
- 8) The final phase of compiler model is _____.
A) Syntax Analysis
B) Code optimization
C) Code Generation
D) Lexical Analysis

Q-2 Write down the answer of following short questions. (Any 7)

[14]

- 1) List all the language processors and explain any one of them.
- 2) What is a role of compiler?
- 3) What is buffer pair?
- 4) What is transition table?
- 5) Differentiate : Lexical analysis vs. Parsing
- 6) What is Parse Tree?
- 7) What is Parser?
- 8) What is symbol system?
- 9) What are the problems generated by the use of registers?

Q-3 A) Explain phases of compiler in detail.

[06]

B) Write a detail note on software productivity tools.

[06]

OR

B) What is language processor? List all the language processors and explain all in detail.

[06]

Q-4 A) Explain the role of lexical analyzer.

[06]

B) Suppose Language L contains those strings which are started with '0' (zero).

[06]

Like $L = \{ 0, 00, 01, 000, 010, 011, 0000, \dots \}$. Convert this expression into NFA, built Transition table and also convert into DFA.

OR

B) What is transition diagram? Write down the rules for constructing transition diagram. Also explain with example.

[06]

Q-5 A) Write a detail note on role of the Parser.

[06]

B) Write a detail note on shift reduce parsing.

[06]

OR

B) Write a detail note on recursive descent parsing.

[06]

Q-6 A) Write a detail note on issues in the design of a code generator.

[06]

B) Explain the principal sources of optimization.

[06]

OR

B) Explain Dead-code elimination and Copy Propagation in detail.

[06]

—X—
(2)